

CHEMICAL DYNAMICS

Orbiting resonances in formaldehyde reveal coupling of roaming, radical, and molecular channels

Casey D. Foley^{1†}, Changjian Xie^{2,3†}, Hua Guo^{3*}, Arthur G. Suits^{1*}

The roaming chemical reaction mechanism involves near-dissociation of an energized molecule to radicals that leads instead to intramolecular reaction after reorientation at long range. Surprisingly, no clear quantum signatures of roaming have been observed to date, despite the quantum nature of the roaming event. We found evidence of quantum dynamics in the photodissociation of formaldehyde near the roaming threshold. This is ascribed to resonances associated to $\text{H}+\text{HCO}(K_a = 1)$ that have a profound impact on the CO rotational and translational energy distributions and cause the roaming fraction to vary by a factor of 2 over an energy range of 10 cm^{-1} . The roaming pathway serves both to modulate and report on the complex vibrational dynamics and coupling among the three dissociation pathways in the excited molecule as it decays to products.

Unimolecular roaming reactions are those in which an isolated molecule undergoes near-dissociation by simple bond fission, but reorientation of the fragments at long range leads instead to an intramolecular reaction (1–3). Roaming reactions, first clearly identified in joint experimental and theoretical studies of formaldehyde photodissociation in 2003 (4), have since become recognized as a universal aspect of chemical reactivity. Experimental studies continue to reveal novel roaming-related phenomena (5, 6), and roaming dynamics have recently been captured in real time (7). Considerable theoretical work is also currently devoted to understanding roaming (8), including from the point of view of transition state theory (9, 10) and dynamical systems (11). With only one exception (12), roaming has been theoretically treated with classical models. No obvious quantum effects in roaming have been reported even though at the roaming threshold in formaldehyde, for example, the H-atom de Broglie wavelength is several angstroms. Nor has evidence emerged to indicate any departure from the ergodic limit for systems undergoing roaming.

Here, we used a full suite of experimental tools combined with theoretical models to show compelling evidence of quantum orbiting resonances at the onset of roaming in formaldehyde. These high-lying metastable quantum states profoundly alter the product internal state distribution and roaming yields in a narrow energy window and reveal interactions between all three dissociation continua.

These quantum states may be viewed as resonances in rotational predissociation to the radical channel (13) that are coupled to the strongly exoergic pathways to molecular products, highlighting the connection of roaming dynamics to cold collision studies where quantum mechanics dominates (14).

Out-of-plane motion and roaming yield

Key features of the formaldehyde potential energy surfaces (PESs) relevant to the present study are highlighted in Fig. 1A. After excitation to the first excited singlet state (S_1), internal conversion to the ground electronic state (S_0) leads to dissociation over a conventional tight transition state (t-TS) to molecular products $\text{H}_2 + \text{CO}$ with characteristic high rotational excitation in CO and low vibrational excitation in H_2 (15). At slightly higher energy, radical products $\text{H} + \text{HCO}$ may be formed, and this onset has an impact on the molecular product channel (16). Beginning near the radical threshold, roaming of one H atom over a loose, nonplanar transition state (17) (r-TS) can also lead to molecular products via abstraction at long range. This abstraction gives highly vibrationally excited $\text{H}_2(v \geq 6)$ with low rotational levels of CO (4). The clear signature of roaming in formaldehyde is thus the formation of low rotational levels of $\text{CO}(v = 0)$ in conjunction with highly vibrationally excited H_2 .

Quantum effects in roaming are most likely to be seen at the radical threshold, and we thus focused our investigation there by exciting to the 2^1_4 band on the S_1 state. We used our ability to tune the excitation energy and angular momentum through selective photoexcitation to perform a detailed examination of the photochemical dynamics for a full range of parent rotational levels from the origin up to *ortho* $J = 4$. As a first step, we recorded and assigned detailed photofragment excitation (PHOFEX) spectra, as shown in fig. S1. We then fixed the pump laser on selected parent

rotational levels and scanned the probe laser to obtain $\text{CO}(v = 0)$ resonance-enhanced multiphoton ionization (REMPI) spectra. The parent rotational state designations do not change in the internal conversion to the highly vibrationally excited molecule on S_0 , although subsequent Coriolis coupling could mix states of different rotational and vibrational symmetries (18). We thus analyzed all of these results with respect to the initial rotational level prepared in S_1 .

In Fig. 1B, we first show raw REMPI spectra obtained after excitation of H_2CO to the 2_{02} and 3_{03} rotational levels, which differ in energy by only 10 cm^{-1} . It is well established that the product CO rotational states below $j_{\text{CO}} = 20$ are largely from roaming events, whereas those above $j_{\text{CO}} = 30$ result almost exclusively from dissociation over the conventional t-TS (19, 20). We observed an obvious large increase in the fraction of molecular products that appeared in roaming-associated product levels ($j_{\text{CO}} < 20$) for the 3_{03} case, largely at the expense of high j_{CO} states associated exclusively with dissociation over the t-TS, although differential branching to the radical channel (discussed below) likely also played a role. This profound modulation of the roaming yield was not seen on the analogous 2^2_4 band, which is 1200 cm^{-1} higher in energy (fig. S4), suggestive of a threshold effect.

To understand this effect, we first turned to quasi-classical trajectory (QCT) calculations (21). All of the full-dimensional theoretical treatments of roaming in H_2CO to date have used QCT methods, and these simulations were key in identifying the roaming mechanism in the first place (8). Newtonian dynamics have been quite successful at reproducing the broad features of the dynamics. Only one such study (22) has examined the effect of parent rotation on formaldehyde decomposition both for the roaming and radical channels, but only a slight suppression of the roaming fraction was found at threshold. Nothing resembling the large effect we observed in the present work was reported, nor have we seen such effects in our own QCT investigations of the effect of parent rotation. Moreover, there were no drastic effects seen in the QCT studies that depend on the initial geometry accessed in S_0 or the total energy, which suggested ergodicity in the dissociation of the highly excited $\text{H}_2\text{CO}(S_0)$. The effects we report here are thus not classical in origin; however, we could use classical trajectories to gain some insight into the associated dynamics. Motivated by consideration of the nonplanar r-TS (Fig. 1A) and trajectory calculations that have consistently shown the importance of orbiting of the roaming H atom about the figure axis in formaldehyde (23), we performed two sets of QCT calculations at the radical threshold. In one case, all the initial positions and momenta were

¹Department of Chemistry, University of Missouri, Columbia, MO 65211, USA. ²Institute of Modern Physics, Shaanxi Key Laboratory for Theoretical Physics Frontiers, Northwest University, Xian, Shaanxi 710127, China. ³Department of Chemistry and Chemical Biology, University of New Mexico, Albuquerque, NM 87131, USA.

*Corresponding author. Email: suitsa@missouri.edu (A.G.S.); hguo@unm.edu (H.G.)

†These authors contributed equally to this work.

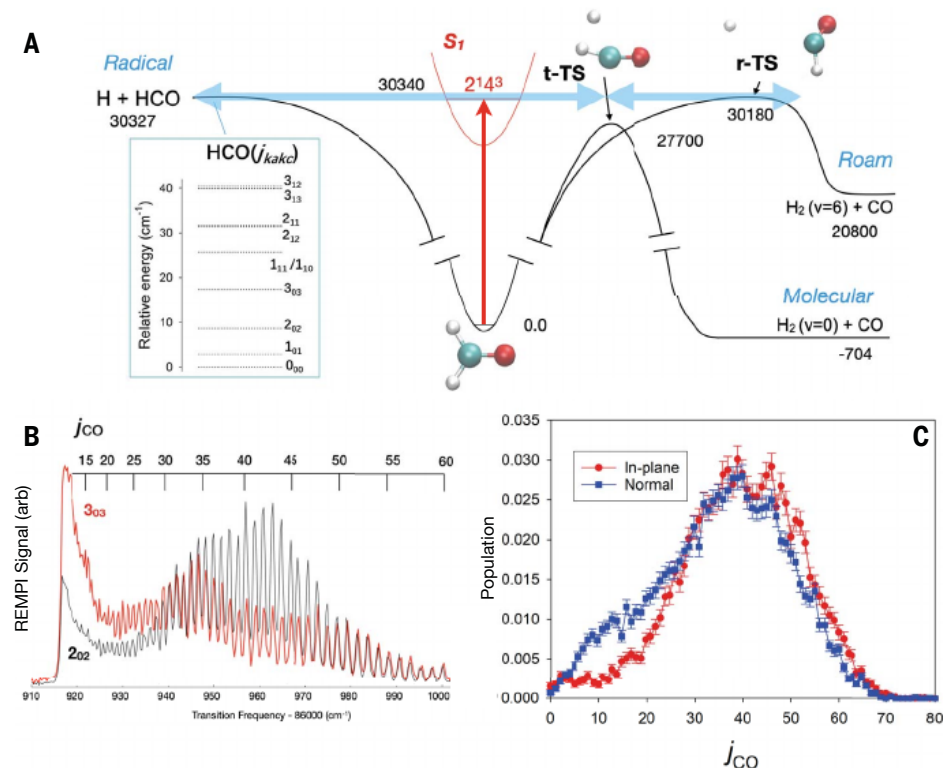


Fig. 1. Roaming and out-of-plane motion in H₂CO. (A) Schematic potential energy surfaces with relevant stationary points showing the three product channels. Energies are in cm⁻¹. (B) CO(v = 0) REMPI spectra for the indicated H₂CO rotational levels in S₁ normalized to area. (C) QCT rotational distributions for CO(v = 0) normalized to area. Error bars denote SE obtained as described in (21).

restricted to the H₂CO molecular plane, whereas in the other no such restriction was imposed. A substantial suppression of the roaming yield (population in $j_{\text{CO}} < 20$) was seen, with a slight relative enhancement of the high- j_{CO} products (Fig. 1C), qualitatively resembling the results in Fig. 1B. The plots in Fig. 1C were normalized to area to facilitate comparison to the experimental spectra in Fig. 1B. The QCT results also showed that the radical channel yield was suppressed for the in-plane restricted case (21). These considerations suggested that the 3₀₃ level involves greater out-of-plane motion than 2₀₂.

Additional insight comes from examining the CO rotational distributions and roaming yield from all parent rotational levels at the onset of the radical channel. These distributions were obtained by fitting REMPI spectra such as those shown in Fig. 1 after accounting for line strengths and degeneracy. These rotational distributions are given in Fig. 2, A and B, grouped according to whether their excitation energy falls within a “resonance window” discussed below. It is obvious that the distributions in Fig. 2B showed a greater yield of roaming (population in $j_{\text{CO}} < 20$) than those in Fig. 2A. We also quantified the roaming fraction (i.e., the roaming yield relative to the total yield of molecular products) that resulted from dissociation for all parent levels

considered (21). The results are shown in Fig. 2D plotted against the energy above the H + HCO threshold (24, 25). We found that levels that fell within an energy window of 5 cm⁻¹ generally showed roaming yields at least 50% greater than those outside this window. Interestingly, the window was just below the asymptotic energy of H + HCO($K_a = 1$) at which free rotation of HCO around its *a* axis is energetically accessible. Furthermore, we found that when the H atom was in the roaming region at 3 to 6 Å, these HCO($K_a = 1$) levels could be stabilized by the H atom, permitting intramolecular orbiting of H around the *a* axis of the incipient HCO. In Fig. 2, we highlight all H₂CO levels below the energy of H + HCO($K_a = 1$) that showed enhanced roaming. This energy window coincides with the average stabilization energy afforded by the H atom at ~5.74 Å from HCO in its equilibrium geometry, according to the PES of the S₀ state. At this distance, the potential in the plane perpendicular to the HCO *a* axis is also quite isotropic, where at shorter distances this is not the case. In addition to 3₀₃, this “resonance window” includes the *para* asymmetry doublets 1₁₀/1₁₁ and 2₁₂/2₁₁. Some of this relative enhancement of roaming yield may also come from suppressed loss of these rotational levels to the radical channel, as discussed below.

Orbiting resonance

The doubling of the roaming fraction in such a narrow energy range was a clear indication that the S₀ dynamics was not always ergodic, as previously believed. It is no surprise that this anomaly occurred so close to the threshold of the radical channel. As we show below, this distinct feature was almost certainly a resonance, or a set of resonances, involving the emergence of orbiting motion of the dissociating H around the incipient HCO that plays a central role in roaming. Access to these orbiting trajectories promotes dissociation via the roaming pathway and likely the radical channel at the expense of the t-TS pathway. Furthermore, the appearance of this resonance profoundly disrupted otherwise largely consistent (and ergodic) dynamics both above and below its onset. This effect may be seen by examining the CO rotational distributions (Fig. 2, A and B). All of the rotational distributions from excitation outside of the resonance window were quite similar in appearance, peaking around $j_{\text{CO}} = 40$, as has been reported for the 2₂₁/2₂₀ *ortho* levels (16). However, the rotational distributions in the vicinity of the resonance showed strikingly different features, with large oscillations and multimodal peaks. The 1₁₀ and 1₁₁ levels, for example, showed complementary maxima and minima within the broad peak that arose from the t-TS pathway, although they differed in energy by only 0.09 cm⁻¹.

Moore and co-workers (26) reported CO rotational distributions for a range of parent rotational levels on the 2¹4¹ band (below the onset of roaming), and these distributions showed variations in the location of the peak from $j_{\text{CO}} = 35$ to 49. These observations were attributed to the chaotic nature of the vibrational wave function projecting on the t-TS, with the configuration of the molecule at the moment of dissociation directly affecting the vibrational excitation of the H₂ fragment and the rotational distribution of the CO fragment. The rotational distributions could be viewed in terms of the reflection principle (27), in which the t-TS nodal structure in the bending mode was projected over the PES onto the product rotational distributions. [Similar nodal structure in the roaming abstraction region has also recently been invoked to account for bimodal CO rotational distributions associated with specific low- j_{H_2} roaming levels (6).] As discussed by van Zee *et al.*, the t-TS nodal structure is determined by the number and character of the channels in S₀ that couple the prepared level in S₁ to the continuum, as well as any interferences among them (26). It was shown that summing over a larger number of channels resulted in broader and smoother rotational distributions, washing out structure that might be seen for some isolated decay channels. Although the orbiting

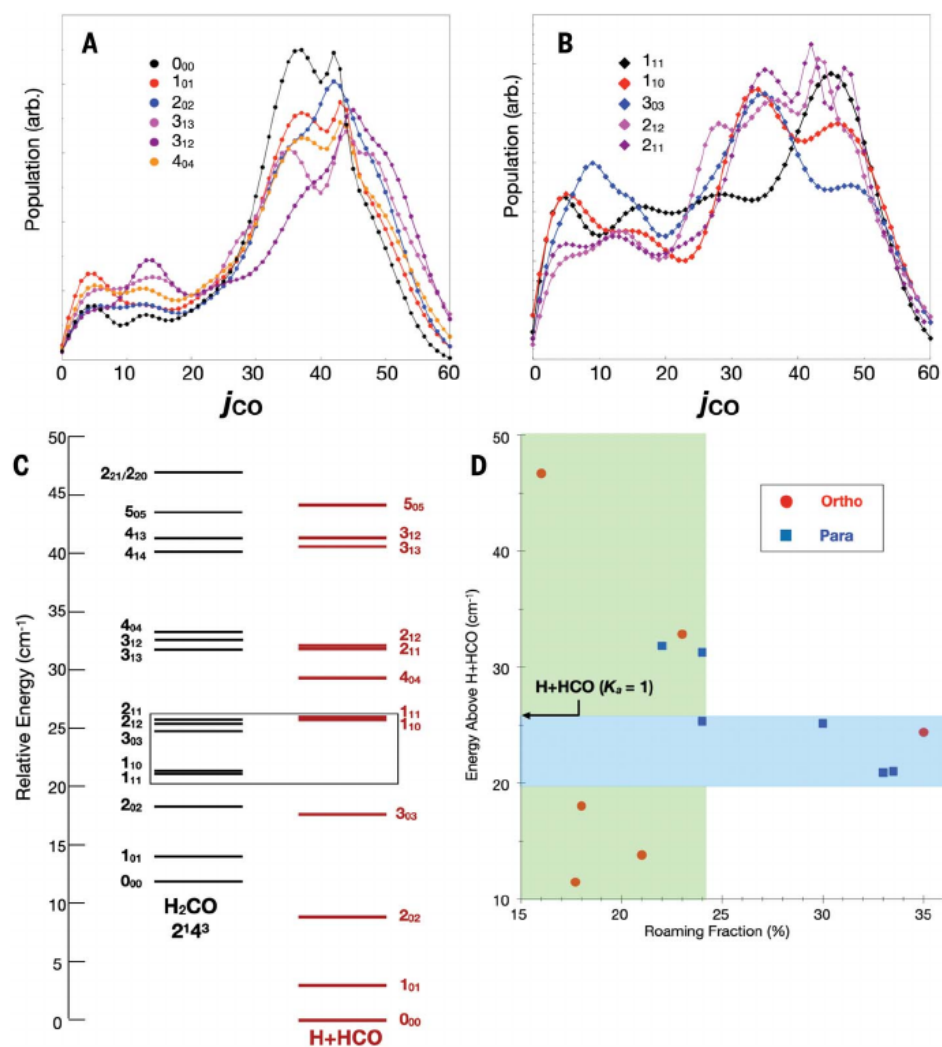


Fig. 2. Evidence of orbiting resonances. (A and B) CO rotational distributions for parent rotational states outside (A) and inside (B) the “resonance window.” Energy values of H₂CO rotational levels in 2⁴₃ relative to H+HCO(J_K, K_c) are adapted from Terentis *et al.* (24). (C and D) Roaming fraction for each probed level (C) versus energy above H+HCO (D).

resonances are fundamentally different from the t-TS “resonances” discussed by van Zee *et al.*, they appear to profoundly distort the flux via the t-TS pathway. We suggest that the impact of the orbiting resonance here was to add a new class of decay channels to S_0 involving out-of-plane motion. Although the total number of decay channels may then grow, which might otherwise have suggested smoother rotational distributions, the fact that they fell into two distinct classes might be the cause of the more structured rotational distributions (and translational energy distributions discussed below) that we observed, which show profound variation with small changes in energy. Other distinct resonances may appear as the following theory suggests, but here we focused on the striking features ~ 23 cm⁻¹ above the radical threshold.

Quantum dynamics in reduced dimensionality

Earlier, we suggested the connection to resonances in rotational predissociation of van der

Waals molecules, and noted that dynamical resonances in scattering (28, 29) and photodissociation (30) are also well known. However, these resonances are observed and characterized theoretically at energies just above the lowest decay channels. We emphasize that this is not the case here: The radical threshold is embedded in the continuum of H₂ + CO at ~ 4 eV above the lowest-energy product channels, so a similarly rigorous treatment of these resonances is currently not feasible. To gain insight into the observed behavior, however, we have performed quantum dynamical calculations in reduced dimensionality, in which the H interacting with HCO at its equilibrium geometry was used in a three-dimensional (3D) quantum model based on the latest *ab initio* PES (37). The three coordinates were the distance between H and the HCO center of mass (R) and the polar (θ) and azimuthal (ϕ) angles related to a body-fixed frame defined by the HCO principal

axes (Fig. 3A) (27). Obviously, such a model is insufficient to describe the full roaming dynamics, but it shed light on the possible signature of resonances near the radical threshold. To this end, we propagated a wave packet with total angular momentum $J = 1$. Note that although the 3D model did not permit the formation of molecular products, it is well established that entering the roaming zone shown in the 3D PES plot in Fig. 3A as a potential well should lead mostly to the roaming products (6). To track the dynamics in the roaming channel, we computed the cross-correlation function with a target function located in the roaming zone. The Fourier transform of the correlation function yielded the energy spectrum with a peak ~ 40 cm⁻¹ above the radical threshold (Fig. 3B), suggesting one or more resonances that were responsible for the enhancement of the roaming channel in the experimental window. Analysis of the scattering wave function (figs. S11 and S12) near the peak energy revealed a notable increase of the out-of-plane component along the a axis of HCO (Fig. 3C), consistent with the hypothesis proposed above. The lower-energy peak with negative amplitude in Fig. 3B may suggest an additional threshold resonance located in the plane, with a different wave function anisotropy. We emphasize that this orbiting resonance is a quantum effect, which cannot be described classically, accounting for the absence of the drastic enhancement of the roaming channel in QCT simulations. Semi-classically, the resonances can be viewed as periodic orbits within the otherwise chaotic classical phase space (17).

Impact on the translational energy distributions

The orbiting resonance effects also had an impact on the total translational energy distributions (TEDs) of the molecular products, which were obtained by velocity map imaging. These distributions are shown in Fig. 4 for selected CO product levels spanning the full range of available states. Here, we compared averaged distributions arising from parent levels within the “resonance window” and those outside, scaled by the associated rotational population [see (27) for the complete set of images and distributions]. Peaks in these state-correlated distributions reflected the associated H₂ vibrational distributions as indicated by the combs above; they could readily be ascribed to the roaming or t-TS pathway, as there was little overlap between them in the asymptotic region. However, it is important to note that some fraction of products that appeared via the t-TS may have also undergone roaming (32), and quantum interference between roaming and nonroaming paths may affect the products that, in the end, arrive via the t-TS and give high j_{CO} .

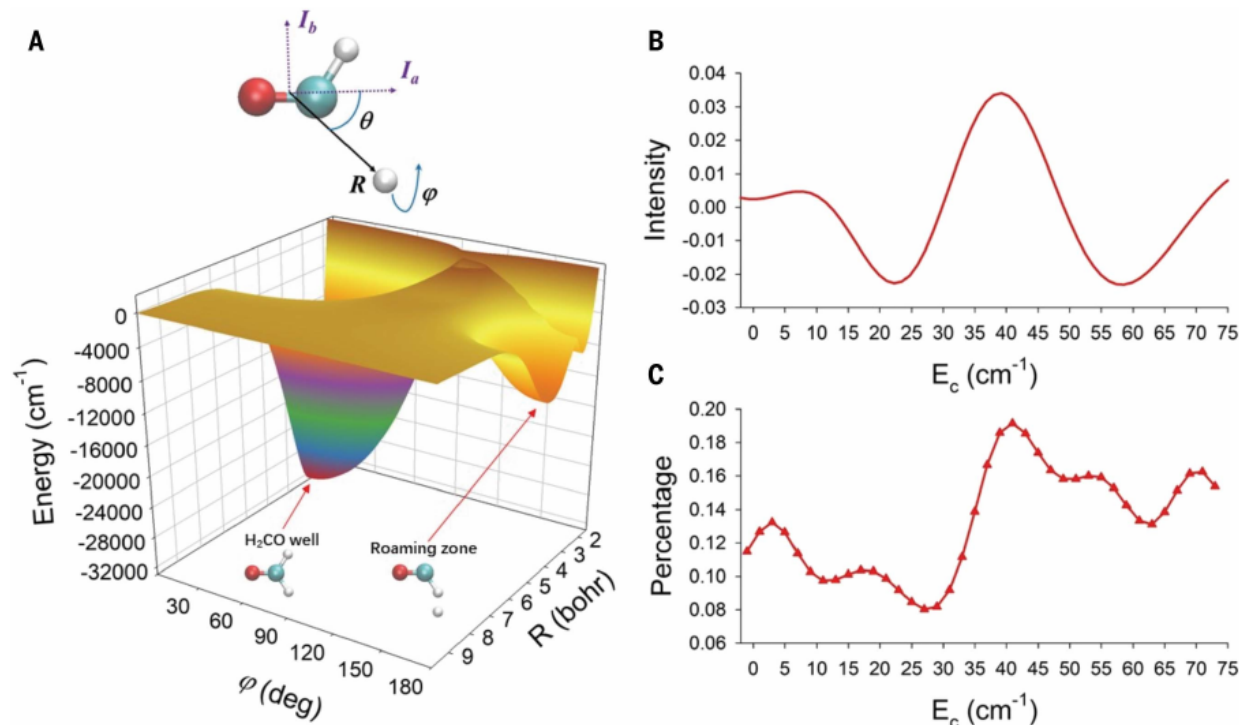


Fig. 3. Quantum dynamics in reduced dimensionality. (A) Scattering coordinates and 3D plot of the PES with θ relaxed, other coordinates fixed at HCO equilibrium geometry, with energy referenced to the H+HCO asymptote (33,087.0 cm⁻¹ above the S_0 minimum. The geometries of the H₂CO well and

roaming zone are shown. (B) The spectrum obtained by the Fourier transformation of a correlation function for H+HCO(1₀₁) with $J = 1$ and $L = 1$. (C) The out-of-plane percentage as a function of collision energy for the scattering wave function in the roaming region $R = 6$ to 9 bohr for H+HCO(1₀₁) with $J = 1$ and $L = 1$.

Figure 4 shows that in general, the low-translational energy (roaming) component was considerably enhanced for the resonance-associated levels, as expected, but the high-translational energy component associated with the t-TS pathway showed large differences as well. For example, in the $j_{CO} = 28$ distribution, the H₂ $v = 0$ to 2 yield was enhanced for the contribution associated to the resonance, the yield $v = 4$ to 5 was reduced, and $v = 3$ was unchanged. In general, it was the higher vibrational levels of the t-TS pathway, $v_{H_2} = 3$ to 5, that were attenuated for the resonance-affected levels. These levels tended to show greater variation in the correlated state distributions at high j_{CO} as well as in the roaming fraction in the range of $j_{CO} = 20$ to 30, where both the roaming and t-TS pathways contributed (fig. S7). Those unaffected by the resonance showed more uniform distributions in this region, and the roaming fraction for each CO product state was reasonably consistent regardless of the parent level excited.

The determination of the roaming yield underlying Fig. 2B came from a combined analysis of the correlated state distributions and the rotational distributions (27). We found that despite some obvious variations, the roaming fraction was largely consistent for a given j_{CO} , and it was the large variation in the relative yield of $j_{CO} < 23$ versus $j_{CO} \geq 23$ that chiefly determined the total roaming

fraction. The orbiting resonance has a large impact on the low j_{CO} yield, shifting selected elements of the high- j_{CO} populations to roaming levels below $j_{CO} = 23$. It is this shuffling of population that alters the rotational distributions for $j_{CO} \geq 23$, hence the distinct appearance of the distributions associated to the resonance in Fig. 2C; the correlated state distributions in Fig. 4 also reflect this impact of the resonance but are also subject to fluctuations associated with the underlying channels coupling S_1 to S_0 even in the absence of orbiting (26).

Although the detailed nature of the correlated state distributions did not directly determine the roaming yield, they provided additional insight into the nature of this shift in population. We saw that $j_{CO} \sim 23$ to 32 preferentially populated H₂ $v = 3$ to 5 on parent levels away from the resonance. A correlation between these intermediate j_{CO} product states and H₂ $v = 3$ was in fact suggested by van Zee *et al.* on the 2¹/₄¹ band, and the analogous effect is seen here on 2¹/₄³ in Fig. 4. We found that this trend was a consequence of excitation in v_5 of the t-TS (see fig. S13): The H-H distance was compressed at the TS and the system began the descent lower along the exit channel. The resonance-associated levels showed an obvious reduction in this contribution, which suggests that these highly strained planar geometries were lost to roam-

ing or to orbiting that ultimately yielded low vibrational levels of H₂ via the t-TS saddle point. Some insight into this impact on the TEDs could be gained by considering the barrierless path joining the out-of-plane roaming saddle point and the t-TS saddle point found by Bowman and co-workers (33). The system undergoing orbiting could “fall” into the region of the t-TS, and the steepest descent path from orbiting at long range would take the system directly to the t-TS, supporting the notion that resonant enhancement of these low- v_{H_2} t-TS levels indeed arose from orbiting as well. Interestingly, this potential anisotropy is the result of a conical intersection between S_1 and S_0 at higher energy (34).

Consequences for the radical channel

The results above underscored the impact of this orbiting resonance on the t-TS pathway and on roaming, but a remaining question concerned the consequences for the radical channel. We had normalized the rotational distributions in Fig. 2 and scaled the TEDs to these rotational distributions; this analysis permitted a rigorous treatment of the relative branching to molecular products but did not account for the impact of possible distinct branching to the radical channel associated with the resonances. It is well established that the radical channel yield rises rapidly in this threshold region. Although we were not able

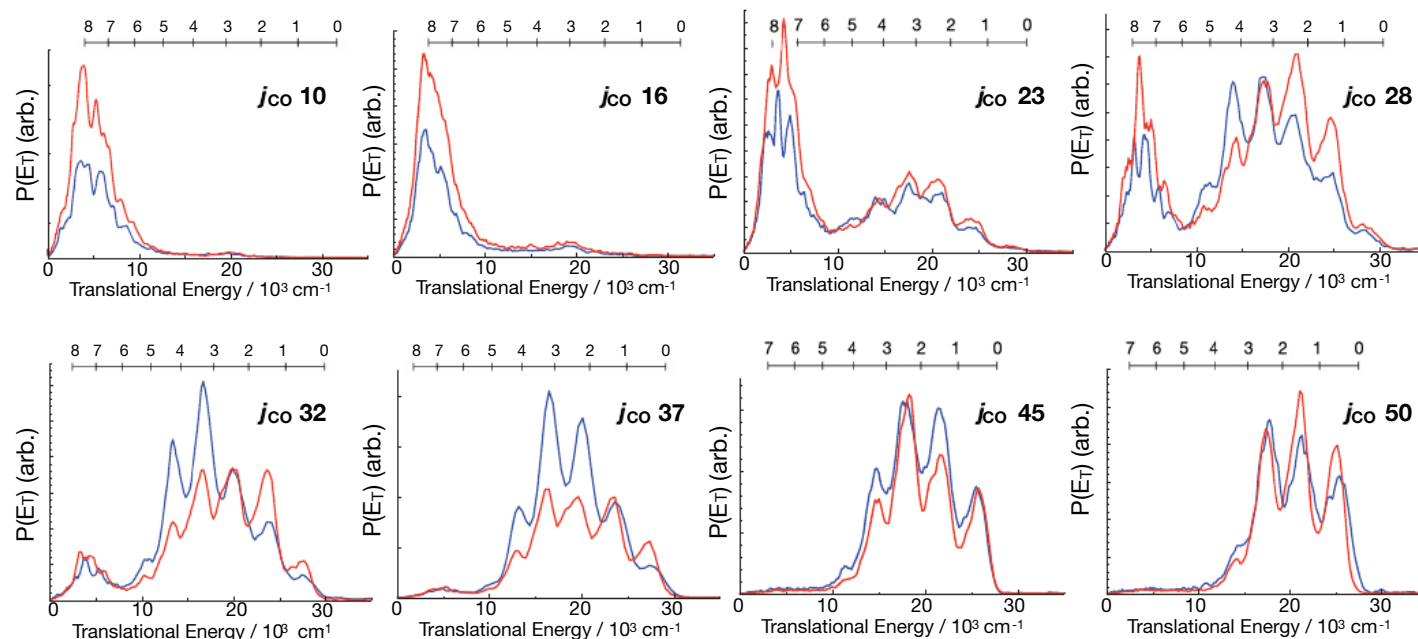


Fig. 4. CO + H₂ translational energy distributions on and off the resonance. Averaged total TEDs obtained after excitation of H₂CO on levels outside of the resonance window (blue line) or within it (red line). The probed CO($v = 0$) rotational levels are indicated in each plot. Combs above each plot show the threshold for forming the indicated vibrational level of the H₂ coproduct. H₂ produced in vibrational levels greater than 5 are associated with roaming. Each distribution has been scaled by the appropriate rotational populations shown in Fig. 2.

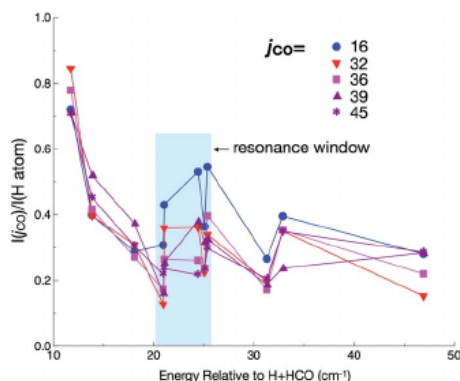


Fig. 5. Product CO ($v = 0, j$) yield relative to the radical channel. The y axis shows the ratio of the integral of the indicated j_{CO} PHOFEX peak to that of the corresponding H atom peak. Blue shading denotes the region of the subject resonance.

to quantify the branching between the radical and molecular channels accurately (21), we could gain some insight using PHOFEX spectroscopy on both H atoms and a range of CO product states.

Figure 5 shows a plot of the ratio of the integrated signal for the indicated CO peak relative to that for the H atom (i.e., the radical channel). We have normalized the two sets of distributions by an average of the signal at the lowest and highest energies farthest from the region of the resonance. We found relatively lower radical branching on the lowest prepared H₂CO levels for all products at the

radical onset, as expected. In the resonance region, we saw striking variations in this trend. Most obvious was the enhancement of $j_{\text{CO}} = 16$ relative to the radical channel. We also found $j_{\text{CO}} = 45$ mildly suppressed. This plot confirmed the impact of the resonance on the radical channel, but additional insight was afforded by fig. S3, in which we plotted the relative branching between $j_{\text{CO}} = 16$, $j_{\text{CO}} = 45$, and H atoms assuming a total quantum yield of 1. We observed attenuation of both the t-TS and the radical channel yield as the roaming fraction grew, which suggests that roaming was capturing flux that would have gone via the radical channel and the t-TS pathway. Kable and co-workers examined HCO rotational distributions after excitation on $2^1 4^3$ at threshold in detail and reported clear deviations from statistical behavior at threshold, consistent with an impact of these resonances (24, 25). Interestingly, levels in the resonance region showed a propensity for production of HCO 2₀₂.

Our results show that the increased orbiting motion of the H atom about the incipient HCO radical just below the opening of the HCO($K_a = 1$) product channel gives rise to greatly enhanced roaming for parent rotational levels within 5 cm⁻¹ of this asymptote. This orbiting also redirects some t-TS flux from strained geometries at the TS, which otherwise would yield excited H₂, to lower TS levels giving colder H₂. It is well known from trajectory calculations that roaming often involves orbiting motion, but in this case the enhanced

roaming possesses a quantum character. When decay to radical products is inhibited, by the small energy deficit in this case, orbiting resonances emerge and redirect product flux from both the t-TS pathway and the radical channel into roaming, and redistribute it within the t-TS pathway as well. These threshold resonance effects underscore the impact of quantum quasi-bound states on the decay dynamics of a prototypical unimolecular reaction.

REFERENCES AND NOTES

1. A. G. Suits, *Acc. Chem. Res.* **41**, 873–881 (2008).
2. J. M. Bowman, B. C. Shepler, *Annu. Rev. Phys. Chem.* **62**, 531–553 (2011).
3. A. G. Suits, *Annu. Rev. Phys. Chem.* **71**, 77–100 (2020).
4. D. Townsend et al., *Science* **306**, 1158–1161 (2004).
5. H. Li, A. Kamasah, S. Matsika, A. G. Suits, *Nat. Chem.* **11**, 123–128 (2019).
6. M. S. Quinn et al., *Science* **369**, 1592–1596 (2020).
7. T. Endo et al., *Science* **370**, 1072–1077 (2020).
8. J. M. Bowman, P. L. Houston, *Chem. Soc. Rev.* **46**, 7615–7624 (2017).
9. S. J. Klippenstein, Y. Georgievskii, L. B. Harding, *J. Phys. Chem. A* **115**, 14370–14381 (2011).
10. D. U. Andrews, S. H. Kable, M. J. T. Jordan, *J. Phys. Chem. A* **117**, 7631–7642 (2013).
11. F. A. L. Mauguère et al., *Annu. Rev. Phys. Chem.* **68**, 499–524 (2017).
12. A. Li, J. Li, H. Guo, *J. Phys. Chem. A* **117**, 5052–5060 (2013).
13. J. M. Hutson, *J. Chem. Phys.* **92**, 157–168 (1990).
14. A. G. Császár et al., *Phys. Chem. Chem. Phys.* **22**, 15081–15104 (2020).
15. C. B. Moore, J. C. Weisshaar, *Annu. Rev. Phys. Chem.* **34**, 525–555 (1983).
16. R. D. van Zee, M. F. Foltz, C. B. Moore, *J. Chem. Phys.* **99**, 1664–1673 (1993).
17. L. B. Harding, S. J. Klippenstein, A. W. Jasper, *Phys. Chem. Chem. Phys.* **9**, 4055–4070 (2007).
18. H. L. Dai, C. L. Korpa, J. L. Kinsey, R. W. Field, *J. Chem. Phys.* **82**, 1688–1701 (1985).

19. S. A. Lahankar *et al.*, *J. Chem. Phys.* **125**, 44303 (2006).
20. M. S. Quinn, D. U. Andrews, K. Nauta, M. J. T. Jordan, S. H. Kable, *J. Chem. Phys.* **147**, 013935 (2017).
21. See supplementary materials.
22. J. D. Farnum, X. Zhang, J. M. Bowman, *J. Chem. Phys.* **126**, 134305 (2007).
23. P. L. Houston *et al.*, *J. Chem. Phys.* **147**, 013936 (2017).
24. A. C. Terentis, S. E. Waugh, G. F. Metha, S. H. Kable, *J. Chem. Phys.* **108**, 3187–3198 (1998).
25. H.-M. Yin, K. Nauta, S. H. Kable, *J. Chem. Phys.* **122**, 194312 (2005).
26. R. D. van Zee, C. D. Pibel, T. J. Butenhoff, C. B. Moore, *J. Chem. Phys.* **97**, 3235–3244 (1992).
27. R. Schinke, V. Engel, *Faraday Discuss.* **82**, 111–124 (1986).
28. T. de Jongh *et al.*, *Science* **368**, 626–630 (2020).
29. K. Liu, *Adv. Chem. Phys.* **149**, 1–46 (2012).
30. S. A. Reid, H. Reisler, *J. Phys. Chem.* **100**, 474–487 (1996).
31. X. Wang, P. L. Houston, J. M. Bowman, *Philos. Trans. R. Soc. A* **375**, 20160194 (2017).
32. P. L. Houston, R. Conte, J. M. Bowman, *J. Phys. Chem. A* **120**, 5103–5114 (2016).
33. B. C. Shepler, Y. Han, J. M. Bowman, *J. Phys. Chem. Lett.* **2**, 834–838 (2011).
34. L. B. Harding, S. J. Klippenstein, A. W. Jasper, *J. Phys. Chem. A* **116**, 6967–6982 (2012).
35. C. D. Foley, C. Xie, H. Guo, A. G. Suits, Zenodo (2021); <http://doi.org/10.5281/zenodo.4975789>.

ACKNOWLEDGMENTS

We thank J. Bowman for sharing the PES used in dynamical calculations, and an anonymous reviewer for useful suggestions. **Funding:** Supported by Army Research Office MURI grants W911NF-19-1-0283 (H.G. and A.G.S.) and W911NF-17-1-0099 (A.G.S.), and by National Natural Science Foundation of China grant 22073073 (C.X.). **Author contributions:** C.D.F.: investigation

(experiment), analysis, writing–reviewing and editing; C.X.: investigation (theory), analysis, writing–reviewing and editing; H.G.: conceptualization, supervision, writing–reviewing and editing; A.G.S.: conceptualization, supervision, analysis, writing–original draft. **Competing interests:** The authors declare no competing interests. **Data and materials availability:** All original data are available at Zenodo (35).

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.abk0634
Materials and Methods
Supplementary Text
Figs. S1 to S13
Tables S1 to S3
References (36–55)

21 June 2021; accepted 4 October 2021
10.1126/science.abk0634